

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

1. Real-Time System Design Trade-off

Question:

A surveillance system must operate under constraints:

- Viola-Jones: Fast but less accurate – **78%**
- YOLO: Moderate speed, good accuracy – **92%**
- Deep Learning: High accuracy – **95%**, but slow

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

Answer:

Viola-Jones provides the fastest performance but has the lowest accuracy of 78%. Deep Learning provides the highest accuracy of 95%, but its slower processing makes it unsuitable when latency is critical. YOLO provides a good balance between speed and accuracy, achieving 92% accuracy with moderate processing speed.

Therefore, **YOLO is the best choice** for a real-time surveillance system. It provides significantly better accuracy than Viola-Jones while maintaining better speed than Deep Learning.

Conclusion:

YOLO should be selected because it offers the best **speed–accuracy trade-off** for real-time surveillance.

2. Robust Detection under Environmental Variations

Question:

Detection accuracy across different environments is:

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions.

Answer:

Viola-Jones performs reasonably well in normal conditions but its accuracy decreases significantly under occlusion and low-light conditions. YOLO performs better, with 90% accuracy in normal conditions, 75% under occlusion, and 70% in low light.

Deep Learning provides the highest accuracy in all three conditions: 95% in normal conditions, 85% under occlusion, and 80% in low light.

Therefore, **Deep Learning is the most robust technique** because it maintains the highest accuracy across all environmental conditions.

Conclusion:

Deep Learning is the best choice when **robustness and accuracy** are more important than computational speed.

3. Motion Estimation vs Computational Cost

Question:

Two motion estimation methods are given:

- Method A: Low error (**5%**), high computation
- Method B: Moderate error (**10%**), low computation

Compare the methods and evaluate their suitability for real-time and offline applications.

Answer:

Method A has a lower error rate of 5%, which means it provides more accurate motion estimation. However, it requires high computational power. Hence, it is more suitable for **offline applications**, where processing time is less important.

Method B has a higher error of 10%, but its computational requirement is low. Therefore, it is more suitable for **real-time applications**, where fast processing is important.

Conclusion:

- **Real-time application → Method B**
- **Offline application → Method A**

The selection depends on whether **speed or accuracy** is the primary requirement.

4. Model Generalization vs Overfitting

Question:

The performance of two models is:

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

Answer:

Model A achieves a very high training accuracy of 98%, but its testing accuracy drops to only 70%. This large difference indicates **overfitting**. The model performs well on training data but does not generalize effectively to unseen data.

Model B has 90% training accuracy and 88% testing accuracy. The small difference between training and testing performance indicates that the model generalizes well to new data.

Therefore, **Model B is more reliable for deployment.**

Conclusion:

Model B should be selected because it has better **generalization and lower overfitting risk**, even though its training accuracy is lower than Model A.

5. Detection Model Comparison under Resource Constraints

Question:

The performance of three models is:

Model	Accuracy	Latency
A	85%	40 ms
B	92%	90 ms
C	90%	60 ms

Compare the models and evaluate which is optimal for real-time systems with moderate accuracy requirements.

Answer:

Model A has the lowest latency of 40 ms, making it the fastest, but its accuracy is only 85%. Model B provides the highest accuracy of 92%, but its latency is also the highest at 90 ms.

Model C provides 90% accuracy with a latency of 60 ms. Since the requirement is a **real-time system with moderate accuracy**, Model C provides a good balance between accuracy and speed.

Comparison:

- Model A → Fastest but lower accuracy
- Model B → Highest accuracy but slower
- Model C → Good accuracy with moderate latency

Conclusion:

Model C is the optimal choice because it provides a balanced combination of **90% accuracy and 60 ms latency**, making it suitable for real-time applications.

Code of Conduct:

I V G ABBIRAMY-192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data

Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application ; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand